

SC-SCOPE quartic normalisation certificate – pinning the Ghat4

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

scscope-floor-sharpening v1.5 left ONE load-bearing residual for the SC-SCOPE endpoint closure: the absolute normalisation of $\widehat{G}^4$ (a factor-2/(2 π)³ convention, flagged as the $M' = -J(0)$ vs $-J(0)/2$ error class). This note pins it and reports the certified closure.

2. The convention is pinned (Parseval)

The quartic-difference contribution uses $\widehat{G}^4 = G * G * G * G = (J * J)(t)$ (quartic-difference-channel v1.0, exact by associativity, $J = G * G$). The only open question is the absolute normalisation of the convolution. By Parseval, with the standard Fourier convention,

$$\widehat{G}^4(0) = (J * J)(0) = \frac{1}{(2\pi)^3} \int J(q)^2 d^3q = \frac{1}{2\pi^2} \int_0^\infty q^2 J(q)^2 dq.$$

Computing the left side by the convolution routine and the right side directly (scscope_quartic_certificate.py),

$$\frac{(J * J)(0)_{\text{conv}}}{(2\pi^2)^{-1} \int q^2 J^2} = 1.0000$$

to four digits. The convolution is therefore STANDARD-normalised; the factor-2/(2 π)³ caveat is resolved. As a cross-check the Young ceiling holds, $(J * J)(0) = 1.27 \times 10^{-3} \leq J(0) \|J\|_1 = 4.76 \times 10^{-3}$ (ratio 0.27), confirming no factor-2 over-count.

3. Certified R_{max} and the certified joint

With the convention pinned, the realized per-transfer ratio

$$R(t) = 12 \left(\frac{5v}{2} \right)^2 \lambda'^{-2} \widehat{G}^4(t) \frac{4(1-a_0)}{J(t)}$$

on the endpoint chords gives $R_{\text{max}} = 0.385 < 0.634$ – CERTIFIED (the Young estimate 1.019 was loose by $\sim 2.6\times$). Under the CONSERVATIVE additive bookkeeping (the corrected one of scscope-floor-sharpening v1.4, NOT the retracted pairing formula),

$$\text{joint} = \frac{\text{MARGIN}}{C_2 + C_{\text{sunset}} + C_{\text{quartic}}}, \quad C_2 = \frac{\text{MARGIN}}{\rho_{\text{lat}}}, \quad C_{\text{sunset}} = \frac{\text{composed}}{1.13}, \quad C_{\text{quartic}} = R_{\text{max}} C_2,$$

$$\text{joint} = \times 1.040 \ (\rho_{\text{lat}} = 6.55) \ \dots \times 1.082 \ (\rho_{\text{lat}} = 12.6) > 1$$

The certified quartic flips the OLD $\times 0.945$ (with the loose $R_{\max} = 1.019$) to closure. The sunset (C_{sunset} , rigorous, M-ENDPOINT $\times 1.13$) is the binding term; the floor sharpening ($\rho_{\text{lat}} \geq 6.55$, proved $K_{\text{floor}} \leq T'$) and the certified quartic together cross unity.

4. Status: certified but THIN – operator re-examination

Every ingredient is now proved/certified: (i) $K_{\text{floor}} \leq T'(M)$ (R-027 $t \neq 0$ part); (ii) the floor $\rho_{\text{lat}} \geq 6.55$; (iii) the sunset $\times 1.13$ (M-ENDPOINT); (iv) $R_{\max} = 0.385$ (this certificate); (v) the conservative additive bookkeeping. The SC-SCOPE all-orders endpoint CLOSES, joint $\geq \times 1.040 > 1$. The closure is THIN ($\times 1.04$ in the worst case $K_{\text{floor}} = T' = n_{\text{pack}}$; $\times 1.08$ with the verified $K_{\text{floor}} \leq 0.52T'$). Per the operator’s instruction (“if $R_{\max} < 0.634$ is certified, then the lift can be re-examined”), this note ENACTS NO lift; it presents the certificate for the operator’s re-examination, given the thin margin. B1 stays T6 on {H-LAYER, SC-SCOPE}.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

- (α) “*The convention could still be off by 2.*” DISMISSED: the Parseval identity $(J * J)(0) = (2\pi^2)^{-1} \int q^2 J^2$ holds to ratio 1.0000; a factor-2 would show as ratio 2 or 0.5. The standard convention is pinned, not assumed.
- (β) “*This repeats the earlier overclaim.*” NO: the earlier lift used the LOCAL pairing formula $\rho/2.872$; this uses the CORRECTED additive bookkeeping (the one that gave $\times 0.945$ before the quartic was certified). The certified quartic is what legitimately flips it, and the result is presented for operator re-examination, not auto-lifted.
- (γ) “ *$\times 1.04$ is too thin to be safe.*” VALID and RECORDED: the closure is thin in the worst case; the verified $K_{\text{floor}} \leq 0.52T'$ gives $\times 1.08$, and any $T' < n_{\text{pack}}$ improves it. The thinness is flagged for the operator; the sign is positive and every input is certified.

Quantitative sanity check. Parseval ratio 1.0000; Young ratio 0.27 (≤ 1); $R_{\max} = 0.385 < 0.634$; joint $\times 1.040$ (cons) / $\times 1.082$ (ver) > 1 ; OLD $R_{\max} = 1.019 \Rightarrow \times 0.945$ (the certification is what flips it). Direction: smaller $R_{\max} \Rightarrow$ smaller $C_{\text{quartic}} \Rightarrow$ larger joint.

6. Result footer

Result ID:	R-029 cont. / SC-SCOPE quartic normalisation certificate
Precise statement:	Ghat4=J*J convolution is standard-normalised (Parseval $(J*J)(0)=(2\pi^2)^{-1} \int q^2 J^2$, ratio 1.0000; Young ratio 0.27). Hence $R_{\max}=0.385<0.634$ CERTIFIED. Conservative additive joint = $\times 1.040$ ($\rho_{\text{lat}}=6.55$) .. $\times 1.082$ ($\rho_{\text{lat}}=12.6$) > 1 : the SC-SCOPE all-orders endpoint CLOSES (THIN; sunset-binding).
Scope:	SC-SCOPE all-orders endpoint (I=2e-3); the corrected (additive) bookkeeping; certified.
Dependencies:	R-027 ($K_{\text{floor}} \leq T'$); proved floor $\rho_{\text{lat}} \geq 6.55$; sunset $\times 1.13$ (M-ENDPOINT); quartic-difference-channel v1.0 (R(t) formula); Parseval (this note).
Evidence grade:	CERTIFICATE: convention pinned (Parseval), $R_{\max} < 0.634$ certified, joint > 1 under conservative bookkeeping

	(THIN). scscope_quartic_certificate.py 5/5.
Reproduction command:	python codes/vacuum/scscope_quartic_certificate.py
Expected output:	5/5; Parseval ratio 1.0000; R_max 0.385<0.634; joint x1.040(cons)..x1.082(ver).
Falsification gate:	Parseval ratio != 1 (factor-2 convention), or a tighter sunset accounting lowering the joint below 1; not observed.
Tier before / after:	NO flip enacted. SC-SCOPE endpoint closure CERTIFIED (thin); presented for operator re-examination of the lift. B1 T6 on {H-LAYER, SC-SCOPE} unchanged; B5 T5.
No-overclaim statement:	The closure is THIN (x1.04 worst case) and the lift is NOT enacted -- it is presented for operator decision per the standing instruction. Every input is certified, but the operator authorises the gate flip.
Next required action:	Operator: re-examine the SC-SCOPE endpoint lift given the certified (thin) closure. If accepted, SC-SCOPE lifts, B1 -> {H-LAYER}.